Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Phase 1: Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

1. Team & Project Overview

Team Name	Monke
Selected Domain	Domain 2
Selected SDG Track (1–17)	SDG 2.1, 2.4, 12.3, 12.5, 13.2
Project / Solution Name	Buns at Work
Name (Leader)	Sean Abraham Summertan
Email & Phone (Leader)	s34n.4.50@gmail.com & +60 17-822-1132
University Affiliation(s)	Xiamen University of Malaysia
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Micro and small bakeries in Malaysia face a critical operational challenge where they rely on guesswork to determine daily production batch sizes. Because busy staff lack the time to manually count leftover stock at closing, owners consistently over-bake to avoid disappointing walk-in customers. This manual approach results in a silent financial bleed and massive preventable food waste. The problem directly affects micro-business owners operating on razor-thin profit margins. Throwing away just RM 50 of bread daily equals RM 1,500 in lost profit every month, a devastating sum that can force a micro-business to close while frontline staff suffer from tedious, error-prone end-of-day stocktaking. This presents critical economic and environmental issues. Economically, it is a survival issue; by replacing guesswork with data-driven batch suggestions, we protect SME profit margins and improve operational productivity (SDG 8). Environmentally, over-baking wastes vital raw ingredients and energy, only for the food to rot in landfills and generate methane. By predicting exact batch reductions, we attack food waste at its source which is by preventing it from ever being baked (SDG 12). Socially and technologically, traditional bakeries are left behind by expensive POS systems. We democratize innovation by bringing advanced AI computer vision to low-tech, traditional *kuih* stalls of Malaysia, empowering them with simple, smartphone-based solutions (SDG 9).

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Buns at Work is a thin, human-in-the-loop decision layer designed specifically for micro and small bakeries. Instead of relying on gut feeling and memory, owners import their current stock batches and messy sales spreadsheets into our platform, which then generates a single, evidence-backed daily list of actions: what to sell first, mark down, donate, or bake tomorrow. It does not replace their existing till or accounting system; rather, it sits on top of it as a smart operational co-pilot. We apply artificial intelligence strictly where it adds value, avoiding the trap of building a generic chatbot. Our core AI is an explainable demand forecaster that analyzes 28-day trailing sales data to learn weekday and seasonal trends, calculating the exact quantity to bake by factoring in shelf-life caps and daypart shares. Crucially, we implement a "data-trust" gate, if historical data is missing, the AI refuses to guess. We also built an

optional GPU-hosted ML challenger, but it must fall back to our reliable rule-based baseline if confidence is low, ensuring the SME never receives a dangerous recommendation. To build this functional MVP, we leveraged key technical skills from the SDC Phase 1 workshops. We utilize cloud hosting via Replit to deploy our Streamlit frontend and FastAPI backend, ensuring a live, accessible demo. Our database architecture relies on SQLite via SQLAlchemy to maintain a robust, immutable First-Expired-First-Out (FEFO) inventory ledger for food safety and human approvals. Finally, we integrate GenAI and ML API concepts by deploying our predictive model to a serverless RunPod GPU endpoint, using FastAPI to handle the contract-first communication and fallback routing between the user interface, our logic engine, and the external ML model.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

To build a functional, deployed prototype of Buns at Work, we plan to use a modern, lightweight stack optimized for speed, explainability, and cloud deployment. For the frontend, we will use Streamlit to rapidly build the interactive dashboard where the bakery manager reviews AI recommendations and clicks to approve or reject actions. The backend relies on Python 3.11+ and FastAPI to handle contract-first communication between the user interface, the database, and external models, running on the Uvicorn ASGI server. Our database architecture uses SQLite via the SQLAlchemy ORM to maintain the First-Expired-First-Out (FEFO) inventory ledger, batch expiry dates, and the immutable audit logs for human approvals, while Pydantic ensures messy CSV imports are safely validated through our data-trust gate. For the artificial intelligence layer, we use a local Rule-Based Forecaster as our reliable default engine, alongside an optional PyTorch and MLflow machine learning challenger trained to predict complex demand quantiles. The entire application is developed and hosted on Replit, while the heavy machine learning inference is deployed to a RunPod serverless GPU endpoint. The simple system architecture flows from the Streamlit user interface to the FastAPI backend, which then queries the SQLite database and routes heavy inference requests to RunPod, with an automatic fallback to the local rule-based engine if the RunPod endpoint fails. Yes, our project utilizes an official SDC sponsor-provided Cloud Platform: RunPod. We use RunPod to host our optional PyTorch Machine Learning Challenger model via a serverless GPU endpoint. Because the PyTorch model requires a GPU to process complex demand forecasting across multiple quantiles, our FastAPI backend sends product data to RunPod and receives the ML predictions, perfectly demonstrating the hackathon's fallback requirement. If RunPod goes down or times out, the system seamlessly reverts to the local rule-based baseline so the bakery manager always receives a reliable recommendation.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We applied the design-thinking methodology through five stages to ensure Buns at Work is viable, user-friendly, and Phase 2-ready. In Empathize, we identified a specific pain point: micro-bakeries rely on memory and spreadsheets, causing wasted food or empty shelves, and lack confidence to learn complex POS systems. In Define, we framed the problem around data trust and operational burden, establishing that

the solution must refuse to guess when data is missing, fit the SME's existing routine without replacing their till, and keep a human in control of every consequential action. During Ideate, we rejected a generic chatbot and designed a thin, human-in-the-loop decision layer generating one daily list of evidence-backed actions, backed by a First-Expired-First-Out ledger and an explainable forecaster. In Prototype, we prioritized resilience using a lightweight stack (Streamlit, FastAPI, SQLite); the optional PyTorch ML challenger runs on RunPod but automatically falls back to a local rule-based baseline, preventing live demo crashes. Finally, during Test, we targeted a first useful recommendation within fifteen minutes and daily reviews under five minutes via plain labels and one-click approvals. Locking MVP scope on Day 1 and a clean-seed demo by Day 4 ensures a robust, Phase 2-ready deployment.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**Target SDG Sub-Target(s)** Our project primarily addresses SDG 12.3, which aims to halve per capita global food waste at the retail and consumer levels and reduce food losses along production and supply chains by 2030. We also contribute to SDG 13.2 by integrating climate measures through the reduction of food waste emissions, and we align with the Domain 2 brief's requirement for SDG 8.2 and 8.3 by improving productivity through technological upgrading and supporting the growth of micro, small, and medium-sized enterprises. **Impact Baseline & KPI** Malaysia generates over 39,000 tonnes of solid waste daily, with food waste comprising approximately 30.6% of the total, and the bakery sector is a significant contributor to this avoidable waste due to overproduction and poor demand forecasting. Published research on Malaysian bakeries indicates that a single bakery can generate between 18.74 kgCO2eq and 878.75 kgCO2eq of food waste-related carbon footprint per week, with waste spikes directly linked to production mismatches during demand fluctuations. Before deploying Buns at Work, we will establish a pre-deployment baseline for a pilot micro-bakery by measuring the percentage of baked goods discarded at end-of-day over a two-to-four week period, tracking the frequency of "sold out" instances for high-demand items during peak hours, and estimating the

Metric Category	Your Response
	kgCO2eq of baseline food waste using published emission factors. Post-deployment, our Phase 2 targets are to reduce daily waste by 30% within four weeks, reduce stockout incidents by 50%, recover 40% of at-risk stock value through approved markdowns, and achieve an estimated 15–25% reduction in CO2e avoided, all measured through system-logged waste entries, POS or manual sold-out logs, and approved markdown actions logged in the system. Our sources include the FAO SDG 12.3.1 Global Food Losses Data Portal, a Malaysian bakery carbon footprint study published in *Applied Mathematics and Computational Intelligence*, Bernama and SWCorp Malaysia 2024 solid waste statistics, and FAO/UN food waste carbon footprint estimates. All baseline and target figures are presented as estimated projections based on published Malaysian bakery waste data and national waste statistics; we will explicitly label these as estimates in our submission, not guaranteed savings, and will validate them during the Phase 2 pilot.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	A typical Malaysian micro-bakery currently discards 5–10 kg of unsold baked goods per day (approximately 10–15% of daily production), while simultaneously experiencing 3–5 stockout incidents per week during peak hours where customers leave empty-handed. At this daily waste rate, a single small bakery generates an estimated 18.74 kgCO2eq to 878.75 kgCO2eq of food waste-related carbon emissions per week, entirely due to overproduction and the absence of demand forecasting, with 0% of at-risk stock currently being recovered through markdowns or donations (all surplus is discarded). These baseline figures are estimated from published Malaysian bakery waste data and will be validated via a 2–4 week audit during the Phase 2 pilot.
Core Measurable KPI *The specific unit you will track*	The specific unit we will track is kilograms of food waste per day (kg/day), measured as the total weight of unsold

Metric Category	Your Response
e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.	baked goods discarded at end-of-day, logged directly within the Buns at Work system when the manager confirms the daily waste entry. This core KPI is supported by three secondary units: stockout incidents per week (counted each time a high-demand item sells out during peak hours), markdown recovery percentage (the value of at-risk stock recovered through approved markdown actions divided by total at-risk stock value), and kilograms of CO2e avoided per week (calculated by multiplying logged waste reduction in kilograms by published FAO emission factors). All four units are derived from the system's immutable activity log, ensuring that every measurement is traceable to a timestamped, human-approved action rather than an estimate.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	The specific unit we will track is kilograms of food waste per day (kg/day), measured as the total weight of unsold baked goods discarded at end-of-day, logged directly within the Buns at Work system when the manager confirms the daily waste entry. This core KPI is supported by three secondary units: stockout incidents per week (counted each time a high-demand item sells out during peak hours), markdown recovery percentage (the value of at-risk stock recovered through approved markdown actions divided by total at-risk stock value), and kilograms of CO2e avoided per week (calculated by multiplying logged waste reduction in kilograms by published FAO emission factors). All four units are derived from the system's immutable activity log, ensuring that every measurement is traceable to a timestamped, human-approved action rather than an estimate.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

✓ Yes ☐ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

✓ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ✓ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



24/10/2026